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CUDA

Cuda Performance fundamentals

Cuda Error Handling:

There could be different errors happening at different parts of the system due to cuda.

Ex) cudaMalloc() error ⇒ Memory not alloc. by GPU.

cudaMemcpy() error ⇒ Invalid pointer / Incorrect Memory transfer

Two categories:

(i) Immediate errors:

→ Memory alloc failure

→ Invalid config due to kernel launch

(ii) Asynchronous execution errors:

→ Errors occuring while GPU is executing & CPU has submitted its work to GPU.

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CPU runs continuously without waiting for GPU work to finish (Asynchronous)

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Error checks:

```
mykernel <<< block, thread >>> ();

cudaError_t err = cudaGetLastError();
if (err != cudaSuccess){
    printf("Kernel launch error: %s \n",
            cudaGetErrorString(err));
}
```

* cudaError_t ⇒ Cuda Defined Datatype used to store error status
* cudaGetLastError ⇒ Retriere last recorded cuda runtime error status
* cudaSuccess ⇒ Successful status.
* cudaGetErrorString(err) ⇒ Turns the error code into Readable message.